

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 16/02/2023

Course:	B. Pharmacy	Sem: VI	
Subject Name:	Pharmaceutical Biotechnology	Subject Code: BP605T	
Max Marks:	75	Duration:	3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)	(10 x 2) = 20
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- i) Write the applications of genetic engineering.
- ii) Enlist various enzymes used in rDNA technique with their functions.
- iii) Differentiate between Major Histocompatibility Complex(MHC) class I and class II molecules.
- iv) Define Immunity. Enlist types of immunity
- v) Give role of microbes in industry.
- vi) Define Antigen and Antibody.
- vii) Give the roles of Southern and Western blotting techniques.
- viii) Define: transduction and conjugation.
- ix) Comment on cloning vectors. Give their examples.
- x) Differentiate between prokaryotic and eukaryotic cells.

Q. 2. Long Answers (Answer 2 out of 3)	(2 x 10) = 20
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- i) What is enzyme immobilization? Describe methods of enzyme immobilization along with their applications.
- ii) Explain in detail about the production and purification of monoclonal antibodies by hybridoma technology. Give their applications.
- iii) Explain the basic principle of rDNA technology. Write a detailed account on human insulin production by rDNA technology.

Q. 3. Short Answers (Answer 7 out of 9)	(7 x 5) = 35
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- i) Write a short note on PCR.
- ii) Explain the types of mutation and mutants with suitable examples.
- iii) Elucidate the structure of immunoglobulins with neat labelled diagram.
- iv) Explain about the production of penicillin by fermentation technique.
- v) Write a note on dried human plasma and plasma substitutes.
- vi) Illustrate large scale production fermenter design with its various controls.
- vii) Describe type I hypersensitivity reaction with suitable diagram.
- viii) Illustrate the working principle and applications of biosensors.
- ix) Elaborate ELISA technique.

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